Customers Like You: Recommendation System Summary

Recommendation

Truncated SVD was selected because the customer-product matrix is highly sparse, and SVD can compress this type of data efficiently.

Results

Compared with a dense representation, using 100 SVD components reduces the matrix from about 189.15 MB to 7.63 MB, a 95.97% reduction. The original sparse matrix itself requires only 4.80 MB, so the main benefit of SVD here is fast recommendation scoring rather than storage alone.

The recommendation system achieved a Hit Rate@10 of 20.60%, compared with 17.28% for the best-seller baseline, an improvement of about 3.3 percentage points. Generating top 10 recommendations for 1,927 customers took 1.56 seconds, showing that the personalized model outperformed the baseline while remaining fast enough for a nightly recommendation process.

PCA produced similar recommendation accuracy, but SVD was preferred because it is better suited to the sparse purchase data.

Compression Loss

Compression reduces thousands of product dimensions to 100 components. This makes the data easier to process, but some information from the original purchase matrix is lost. At 100 components, the SVD model had a relative reconstruction error of 0.4961.

The goal is not to reproduce every purchase exactly, but to preserve enough of the underlying purchasing patterns to make useful recommendations.

Customer Groups

The analysis identified four broad customer segments:

Home & Utility	2,375 customers
Novelty & Party	1,575 customers
Decorative Homeware	831 customers
Bags & Storage	693 customers

These groups give the marketing team a simple way to target campaigns around different purchasing patterns.

Risks and Next Steps

The system will be less effective for new customers with little or no purchase history, and recommendations for rare products may be weaker. Customer behavior may also change over time, so recommendation quality should be monitored against the best-seller baseline.

I recommend testing the SVD recommender as a pilot, refreshing recommendations regularly, and tracking Hit Rate@10 to confirm that personalization continues to outperform the simple best-seller approach.